

Report 02-19-2026

Daily Learning Report

Topic: Microsoft Defender XDR (Day 3)

Date: 19/02/2026

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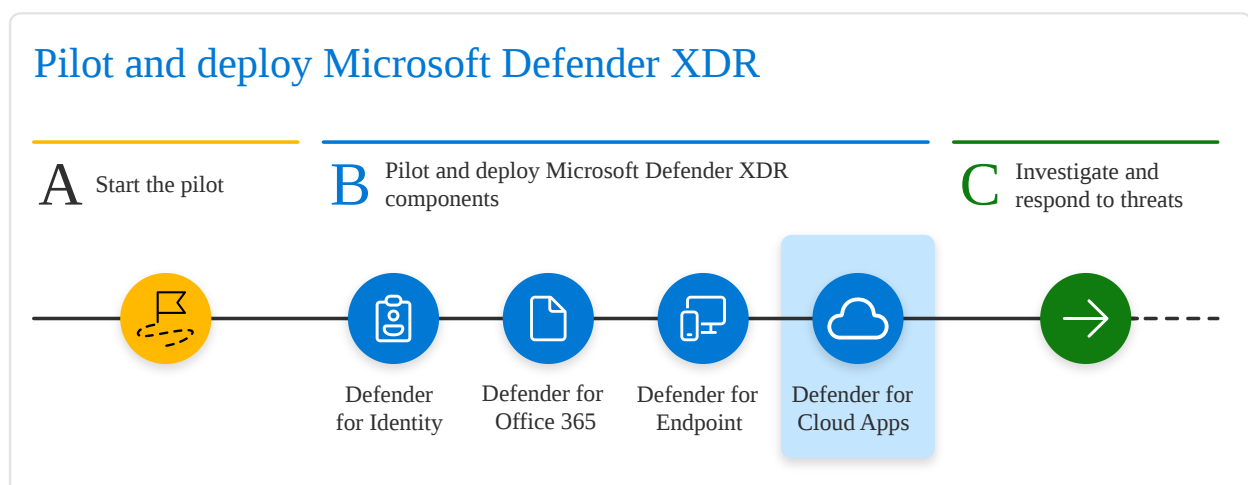
Note: For a better reading experience, please refer to the full version of this report on [Notion](#)

Goal of the Day

- Continue reading about Piloting
- Get started with Microsoft XDR

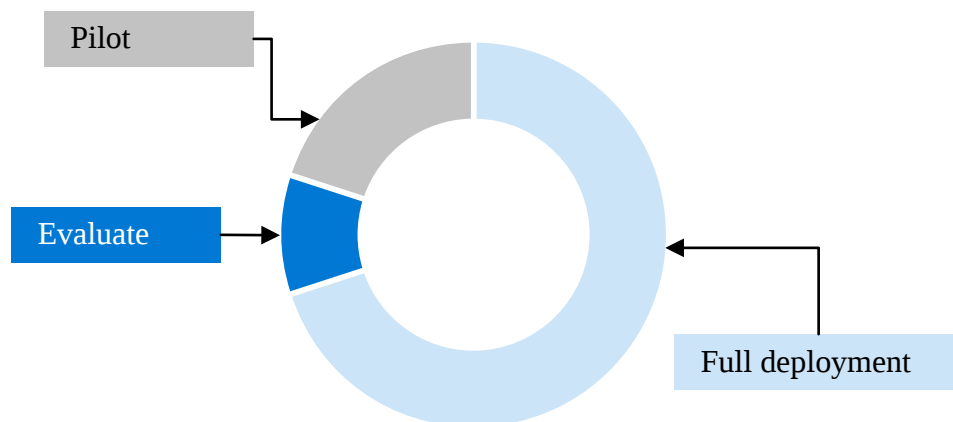
What I have learned today:

I.Pilot and deploy Microsoft Defender for Cloud Apps:



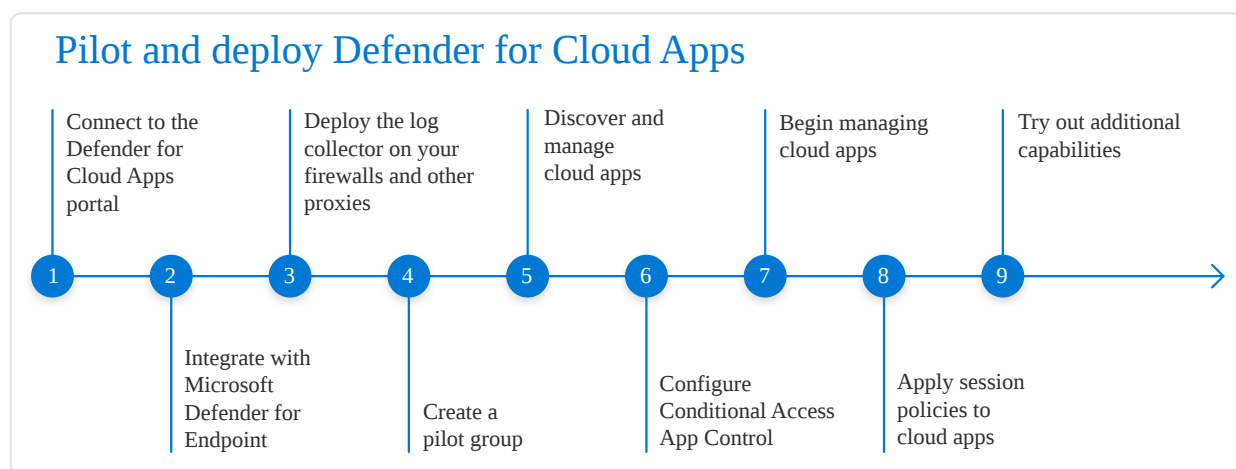
*Pilot and deploy workflow for Defender for Cloud Apps:

The following diagram illustrates a common process to deploy a product or service in an IT environment.



Simply we start by evaluating the product/service , pilote it by a subset of a production environment for testing , then we start deploying the infrastructure gradually .

Heres the workflow provided by microsoft for piloting and deploying Microsoft Defender for Cloud



Follow these steps:

1. Connect to Defender for Cloud Apps
2. Integrate with Microsoft Defender for Endpoint
3. Deploy the log collector on your firewalls and other proxies
4. Create a pilot group
5. Discover and manage cloud apps
6. Configure conditional access app control
7. Apply session policies to cloud apps
8. Try out additional capabilities

Protecting your organization from hackers:

Defender for Cloud Apps detects abnormal user and app behavior, such as impossible travel and unusual data access, and helps prevent lateral movement and data exfiltration.

Defender for Cloud Apps role as a CASB

CASB is a cloud security tool that monitors and controls access to cloud applications and helps detect and prevent attack vectors and data risks.

Microsoft Defender for Cloud Apps delivers full protection for SaaS applications, helping us monitor and protect our cloud app data across the following feature areas:

1. **Fundamental cloud access security broker (CASB) functionality:** monitors and controls how users access cloud (SaaS) applications .It helps discover **Shadow IT**, gives visibility into cloud app usage, and protects data from app-based threats and policy violations.
2. **SaaS Security Posture Management (SSPM):**SSPM helps organizations **improve the security configuration** of their SaaS apps (like Microsoft 365, Salesforce,) by identifying misconfigurations, risky settings, and providing security recommendations.

3. Advanced Threat Protection: Enabling the correlation between the different Microsoft Defender XDR
4. **App-to-app protection:** This protects against risks from **OAuth-enabled apps** that are granted permissions to access cloud data.

Explaining the steps in details

Step 1. Connect to Defender for Cloud Apps

-Start by verifying licenses then connect to Defender for Cloud Apps

Step 2: Integrate with Microsoft Defender for Endpoint

-This step means **connecting** Microsoft Defender for Cloud Apps with Microsoft Defender for Endpoint.

- Defender for Endpoint already **sees device activity** on Windows 10/11 machines.
- When you turn on the integration (just a **toggle** in Microsoft Defender XDR):
 - Cloud Apps can **see which cloud apps users access from their devices**
 - Discovery works **inside and outside the corporate network**
 - Security teams can **investigate cloud app usage per device**
- The results appear in the **Cloud Discovery dashboard** with richer visibility.

Step 3: Deploy the Defender for Cloud Apps log collector on your firewalls and other proxies

Defender for Cloud Apps can be integrated at the network level by deploying a log collector on firewalls and web proxies. This allows the solution to collect internet traffic logs from all devices connected to the network and send them to Defender for Cloud Apps for analysis. Using this data, organizations can discover cloud applications in use, identify risky or unsanctioned apps, and gain visibility even for unmanaged or non-Windows devices.

Defender for Cloud Apps can directly connect to popular cloud applications (like Microsoft 365, Salesforce, Google Workspace, etc.) using their APIs.

These built-in app connectors allow Defender for Cloud Apps to see user activity, monitor data usage, detect risky behavior, and enforce security controls inside the apps themselves—not just network traffic.

In short, API connectors give deep visibility and control over cloud apps without installing agents on devices.

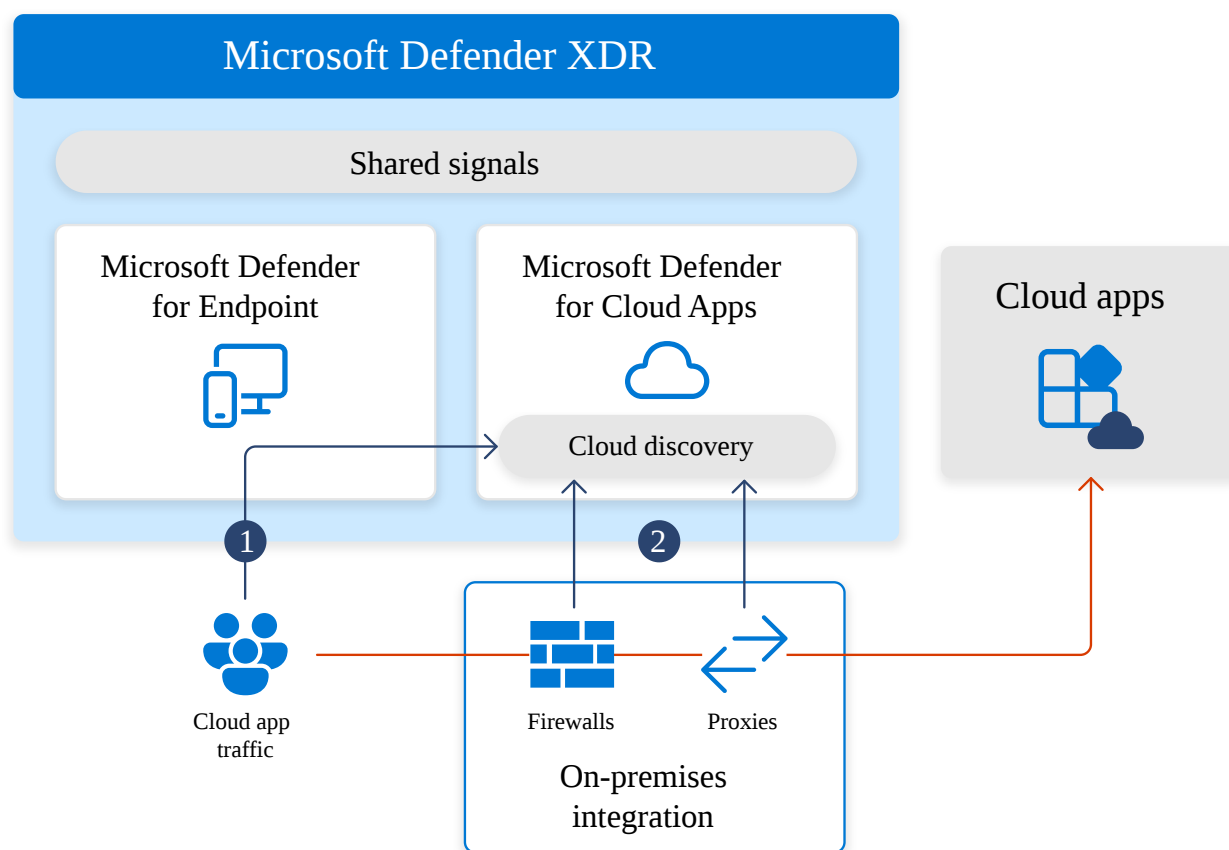
Step 4. Create a pilot group — Scope your pilot deployment to certain user groups

In this step we can choose who to monitor and who to exclude from monitoring .

Step 5. Discover and manage cloud apps:

In order to apply protection we should be knowing about all the cloud apps in the organization

Discover cloud apps



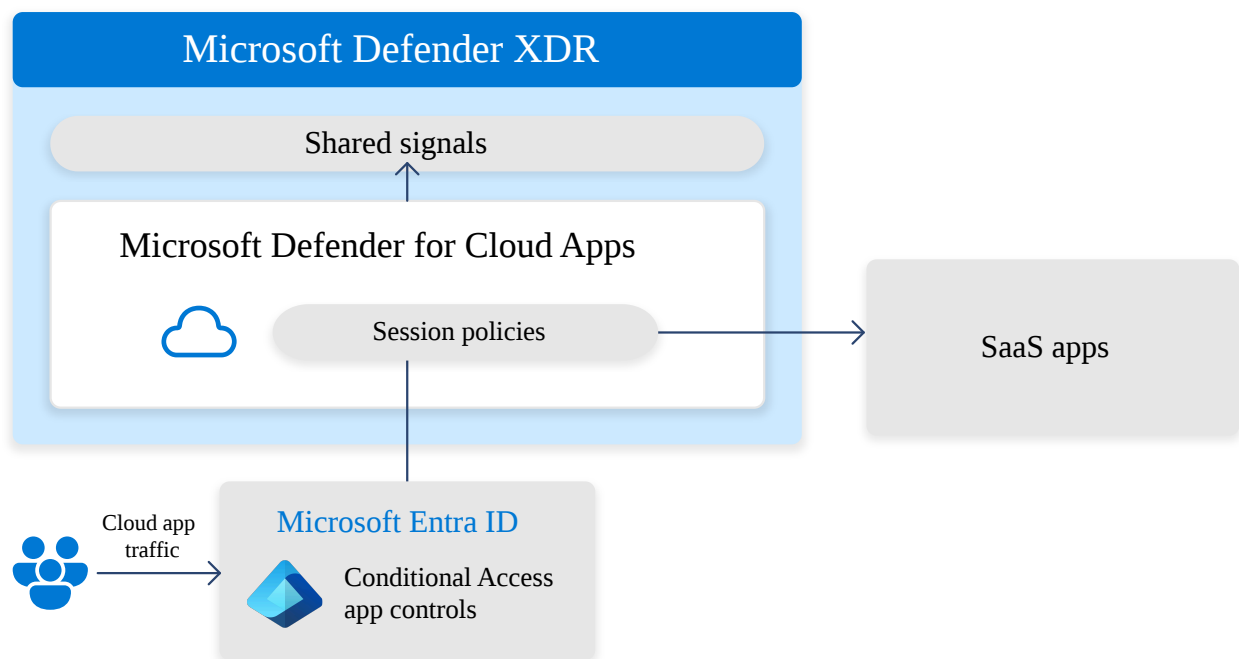
two methods to know the apps running :

1. Cloud App Discovery integrates with Microsoft Defender for Endpoint natively.
2. For coverage on all devices connected to a network, you install the Defender for Cloud Apps log collector on firewalls and other proxies to collect data from endpoints. The collector sends this data to Defender for Cloud Apps for analysis.

-View the Cloud Discovery dashboard to see what apps are being used in your organization.

Step 6. Configure conditional access app control

Conditional Access App Control in Defender for Cloud Apps lets us route traffic from managed SaaS apps through the service to monitor activity and enforce policies like device health and session controls via Microsoft Entra ID integration.



1. **SaaS apps integration** :SaaS apps are connected to Microsoft Entra tenant.
2. **Conditional Access enforcement** : Microsoft Entra ID enforces policies like multifactor authentication for these apps.
3. **Traffic routing policy** : A policy in Microsoft Entra ID to specify which apps we apply the policies to .

4. **Monitoring and session control** : Defender for Cloud Apps inspects the traffic and applies session controls set by administrators.

Step 7. Apply session policies to cloud apps

In this step, we apply session policies to our sanctioned cloud apps using Microsoft Defender for Cloud Apps. These policies control how users and devices access the apps:

- Access from users and devices is routed through Defender for Cloud Apps.
- Only **sanctioned apps** are affected; unsanctioned apps are not controlled.
- Session policies let **us set rules**, for example: allow only managed devices, or monitor traffic from unmanaged devices to evaluate risk.

Step 8. Try out additional capabilities

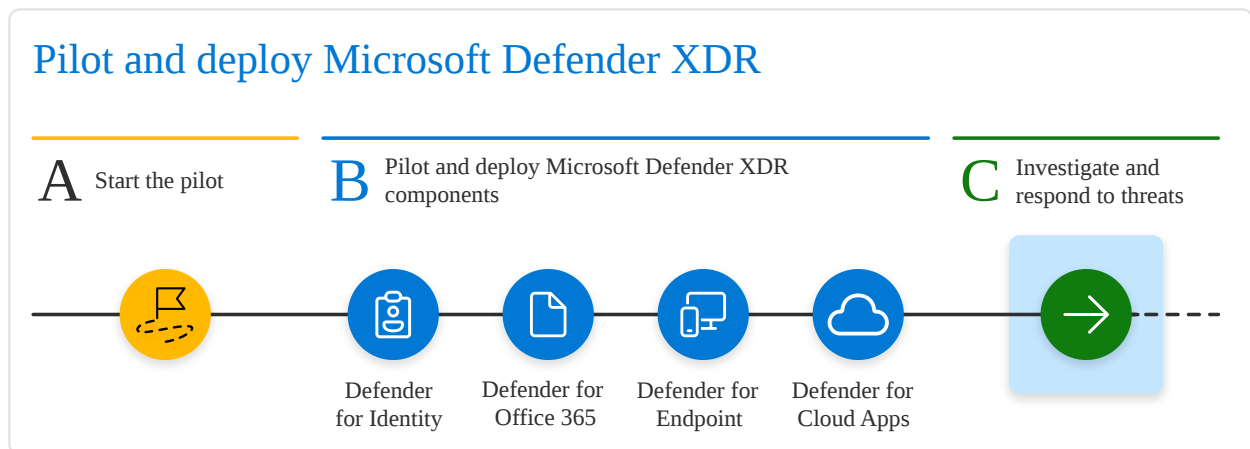
We Can check those for more capabilities ``

- Detect suspicious user activity
- Investigate risky users
- Investigate risky OAuth apps
- Discover and protect sensitive information
- Protect any app in your organization in real time
- Block downloads of sensitive information
- Protect your files with admin quarantine
- Require step-up authentication upon risky action``

II. Investigate and respond using Microsoft Defender XDR

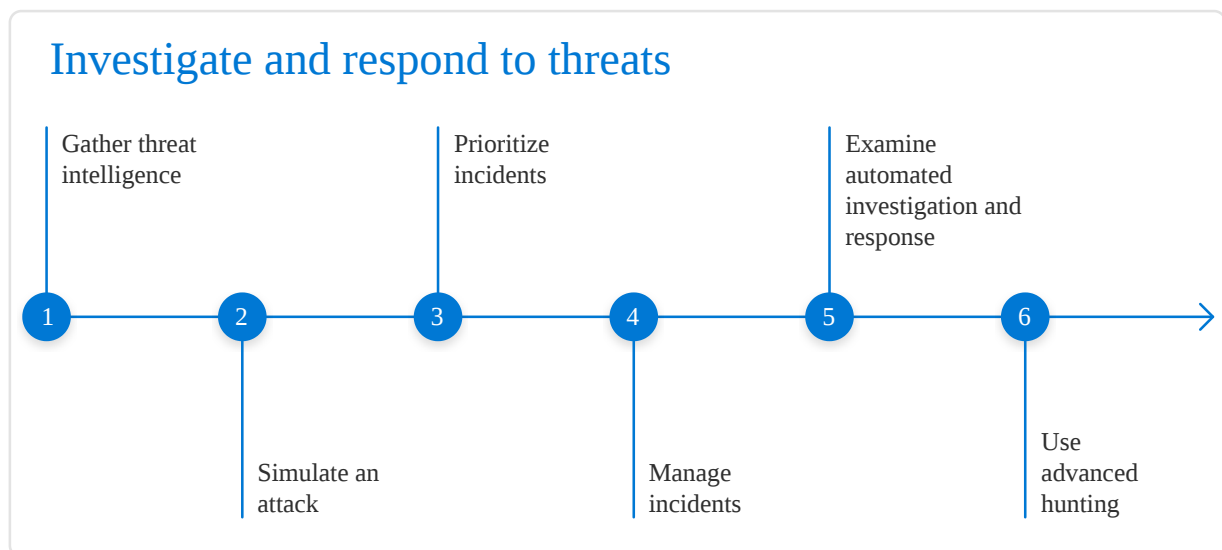
In Microsoft Defender XDR, an alert is a notification generated by Microsoft 365 services when a suspicious or malicious activity is detected. Multiple related alerts are correlated into an incident, which provides a complete view of an attack across devices, users, and mailboxes, helping us investigate and respond effectively.

End-to-end deployment for Microsoft Defender XDR



Workflow for incident investigation and response using Microsoft Defender XDR

Here's the workflow for investigating and responding to incidents using Microsoft Defender XDR in a production environment.



Follow these steps:

1. Gather threat intelligence to identify adversaries and their infrastructure
2. Simulate attacks with the Microsoft Defender portal
3. Prioritize incidents
4. Manage incidents
5. Examine automated investigation and response with the Action center
6. Use advanced hunting

Step 1. Gather threat intelligence to identify adversaries and their infrastructure:

Microsoft Defender Threat Intelligence provides up-to-date threat data, including indicators of compromise (IOCs), threat actors, and vulnerabilities, sourced from Microsoft experts. We can use it to research threats and guide investigations or responses.

Step 2. Simulate attacks with the Microsoft Defender portal:

Microsoft has built in capabilities to perform simulated attacks .

more info exist here :

Simulate an attack with an isolated domain controller and client device (optional).

Step 3. Prioritize incidents

We get to the incident queue from **Incidents & alerts > Incidents** on the quick launch of the Microsoft Defender portal

To examine the list of incidents we can :

- Configure customizable columns
- Apply filtering

Step 4. Manage incidents

Manage incident can you be used to manage accidents.

Here's an example.

Incidents > M365D Demo 06-30-2022 Multi-stage incident involving initial access & Collection on one endpoint reported by multiple sources

M365D Demo 06-30-2022 Multi-stage incident involving Init...

Attack story

Alerts (13)

Devices (1)

Users (2)

Mailboxes (1)

Apps (6)

Investigations (1)

Evidence and Response (25)

Summary

Alerts

1/13 Active alerts

Unpin all

Show all

Incident graph

Layout

Group similar nodes

Jun 30, 2022 10:09 AM

Resolved

Suspicious sequence of exploration activities

workstation1.secop.ninja 2 Users

Jun 30, 2022 10:09 AM

Resolved

Suspicious URL clicked

workstation1.secop.ninja K Dickens@secop.ninja

Jun 30, 2022 10:09 AM

Resolved

User accessed link in ZAP- quarantined email

K Dickens

Jun 30, 2022 10:10 AM

Resolved

A potentially malicious URL click was detected

Karl Dickens

Jun 30, 2022 10:10 AM

Resolved

Suspicious System Owner/User Discovery

workstation1.secop.ninja 2 Users

Jun 30, 2022 10:10 AM

Resolved

Suspicious behavior by svchost.exe was observed

workstation1.secop.ninja K Dickens

Jun 30, 2022 10:11 AM

Resolved

A process was injected with potentially malicious code

workstation1.secop.ninja K Dickens

Jun 30, 2022 10:11 AM

Resolved

Suspicious System Hardware Discovery

workstation1.secop.ninja 2 Users

Incident graph

Layout

Group similar nodes

4 Processes

workstation1

stefgato.companypayroll.ico

MICROSOFT

K Dickens@secop.ninja

Communication

Association

Manage incident

Incident name

M365D Demo 06-30-2022 Multi-stage incident involving Initial access & Collection on ...

Incident tags

Type to find or create tags

Assign to

tamuto@secopninja.onmicrosoft.com

Status

Resolved

Classification

Not set

Comment

Add comment

Save

Cancel

Step 5. Examine automated investigation and response with the Action center

We can see all the action whether pending or listing in the Action Center .

Here's an example.

Home

Exposure management

Investigation & response

Incidents & alerts

Hunting

Actions & submissions

Action center

Submissions

Action Center

Email notifications

Pending

History

Export

2 items

Customize columns

Filter

☐	Action update time ↓	Investigation ID	Approval ID	Action type	Details	Entity type
☐	Apr 23, 2025 3:03 PM	24c913		Soft delete emails	BodyFingerprintBin1("201193991") and P2SenderDomain("v...	Email Cluster
☐	Apr 23, 2025 3:03 PM	2964bc		Soft delete emails	BodyFingerprintBin1("201193991") and P2SenderDomain("v...	Email Cluster

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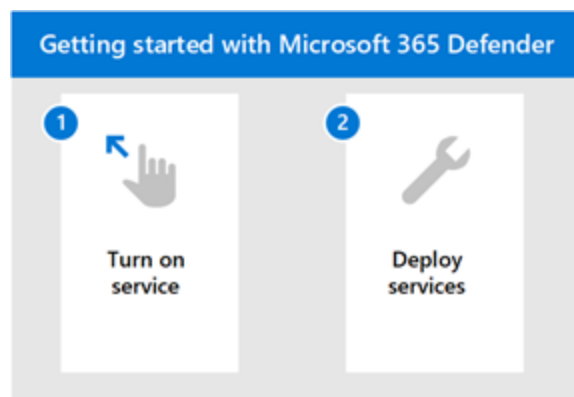
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Step 6. Use advanced hunting

To summarize this part we use Microsoft Defender XDR's **Advanced Hunting** to proactively search through alerts, email events, and endpoint data for suspicious activities. This allows us to simulate attacks, investigate attachments, track file activity on devices, and create **custom detection rules** that automatically generate alerts for future incidents.

I. Get started with Microsoft Defender XDR

Microsoft Defender XDR is a unified experience where we can monitor and manage security across the enterprise. With the integrated alerts across identities, endpoints, data, apps, email, and collaboration tools - investigating and responding to threats now happen in a central location.



General steps to get started

Step 1: Turn on Microsoft Defender XDR (Incoming explanation)

Step 2: Deploy supported services

Deploying services effectively increases the visibility across the network

Key Capabilities (Summary for memorizing) :

Capability	Description
Microsoft Defender for Endpoint	Endpoint protection suite built around powerful behavioral sensors, cloud analytics, and threat intelligence
Microsoft Defender for Office 365	Advanced protection for the organization's apps and data in Office 365, including email and other tools
Microsoft Defender for Identity	Defend against advanced threats, compromised identities, and malicious insiders using correlated Active Directory signals
Microsoft Defender for Cloud Apps	Identify and combat cyberthreats across Microsoft and third-party cloud services

II.How to turn on Microsoft Defender XDR ?

1.Turn on Microsoft Defender XDR

In order to turn on Microsoft Defender we have to start by checking license eligibility and required permissions

Check license eligibility and required permissions

A license to a Microsoft 365 security generally is enough to get Microsoft Defender XDR . It's recommended that we get a Microsoft 365 E5, E5 Security, A5, or A5 Security license or a valid combination of licenses that provides access to all supported services.

Microsoft lists several licenses that include XDR access at no additional cost. These include:

- **Microsoft 365 E5 or A5**
- **Microsoft 365 E3 with Microsoft Defender Suite add-on**
- **Microsoft 365 E3 with EMS (Enterprise Mobility + Security) E5 add-on**
- **Microsoft 365 A3 with A5 Security add-on**
- **Windows 10 or 11 Enterprise E5/A5**
- **Office 365 E5 or A5**
- **Microsoft Defender for Endpoint**

- **Microsoft Defender for Identity**
- **Microsoft Defender for Cloud Apps**
- **Microsoft Defender for Office 365 (Plan 2)**
- **Microsoft 365 Business Premium**
- **Microsoft Defender for Business**

Check your role

to turn on Microsoft Defender we must be one of those roles

- **Global Administrator** (*full access to all Microsoft 365 and Azure settings*)
- **Security Administrator** (*manage security policies and settings across Defender and security features*)
- **Security Operator** (*view and respond to active security alerts and threats*)
- **Global Reader** (*read-only access to all Microsoft 365 and Azure settings and reports*)
- **Security Reader** (*read-only view of security reports and alerts*)
- **Compliance Administrator** (*manage compliance policies, eDiscovery, and governance*)
- **Compliance Data Administrator** (*track and manage compliance data and alerts*)
- **Application Administrator** (*manage enterprise app registrations and application settings*)
- **Cloud Application Administrator** (*manage cloud app security settings and policies*)

Configuring the firewall:

To ensure the Microsoft Defender security services work smoothly — especially when accessing the **Microsoft Defender portal (<https://security.microsoft.com>)** — we need to configure our **network firewall** to allow certain outbound traffic. This includes permitting the necessary IP addresses and domains that Microsoft services use, such as those used by **Azure Monitor** and other Defender services. Firewall allowlisting ensures that services like Defender for Endpoint, Office 365

protection, Defender for Identity, and Defender for Cloud Apps can communicate properly with Microsoft's cloud infrastructure.

Onboard to the service:*Onboarding to Microsoft Defender XDR begins automatically when we access any core feature in the security portal.

Data center location:

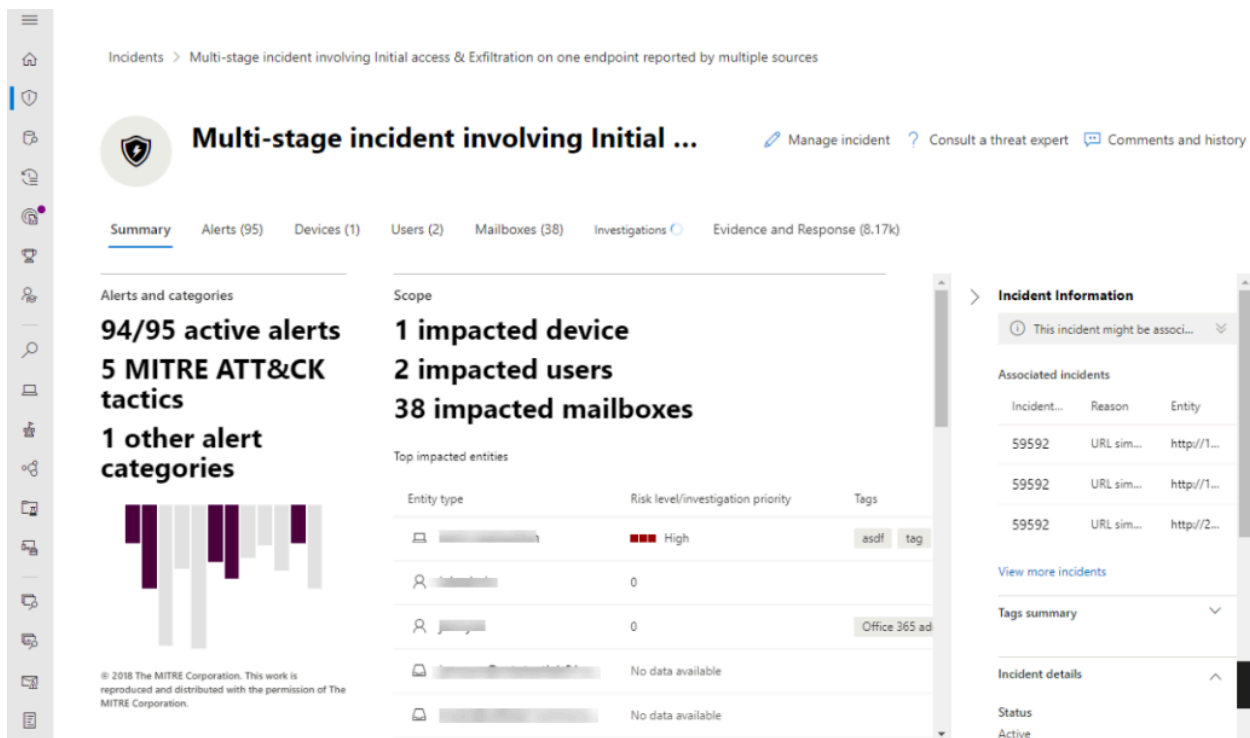
Microsoft Defender XDR uses the **same data location** as Microsoft Defender for Endpoint to store and process security data. If Defender for Endpoint is **not already deployed**, Microsoft automatically chooses a **data center location** based on where your Microsoft 365 security services are active. This ensures data stays in the correct regional location and follows Microsoft's data residency rules.

Confirm that the service is on

Once the services are provisioned it adds :

- Incidents management
- Alerts queue
- An action center for managing automated investigation and response
- Advanced hunting capabilities
- Threat analytics

Microsoft Defender portal with incidents management and other capabilities



Note: To activate mdi i have to at least log in MDC apps once

I have finished the day by understanding the differences between provision and on-boarding and heres a very simple example :

Provisioning VS On-Boarding

Use case: Setting up Microsoft Defender XDR

1.Provisioning (making the service available)

We buy or already have a Microsoft 365 security license (like E5).

Once the license is assigned, **Microsoft Defender XDR is provisioned automatically.**

This means the Defender XDR features (incidents, alerts, hunting, threat analytics) appear in the security portal, but **no data is coming in yet.**

→ At this stage, the system is **ready but empty.**

2. Onboarding (starting to use the service)

we **onboard services and devices**:

- We onboard endpoints using Microsoft Defender for Endpoint
- We connect email using Defender for Office 365
- We integrate identity and cloud apps

Now, devices and services start **sending security signals** to Defender XDR.

→ At this stage, we can **see alerts, incidents, and investigations.**

In simple Words :

Provisioning is turning the service ON

On-boarding is connectig stuff to it